

Aarush Kandukoori

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Education

Carnegie Mellon University

B.S. Computer Science, Statistics & Machine Learning

Pittsburgh, PA

Aug. 2025 – May 2028

Coursework: Imperative Computation, Functional Programming, Fundamentals of Programming, Probability Theory, Statistics & Data, Art & Machine Learning

Experience

BUMP (bump-labs.com)

Pittsburgh, PA

Co-Founder & Chief Executive Officer

Aug. 2024 – Present

- Lead a 3-person team building a patent-pending closed-loop wearable that detects sustained bradycardia and autonomously delivers atropine (USPTO provisional #63/686,176), advised by Harvard Medical School faculty.
- Architected the real-time detection stack: **five Dockerized services** over **Redis Streams** into **TimescaleDB**, a **FastAPI** REST/WebSocket layer, and a React and Vite dashboard, all gated on **GitHub Actions** CI.
- Implemented a streaming Pan-Tompkins QRS detector validated at **0.9996 sensitivity**, **1.00 PPV** on MIT-BIH annotations, and trained a 1D-CNN beat classifier in **PyTorch** exported to **ONNX**, served at **0.06 ms/beat** to model on-device inference.
- Enforced a **250 ms** sensor-to-alert latency budget via end-to-end integration tests and **Prometheus** histograms; validated the Arduino and peristaltic-pump hardware loop at **92.77%** dosing accuracy.

Carnegie Mellon University, Database Group

Pittsburgh, PA

Undergraduate Researcher, Query Engines & Storage

June 2026 – Present

- Extending a published vertical-partitioning result for Apache Parquet from TPC-H to the **99-query TPC-DS** suite, measuring whether splitting a table's columns across separate files is free at query time under **Apache DataFusion**.
- Modified the **Rust** benchmark harness to register single-file, 2-way, and 3-way partitioned variants across all TPC-DS tables and time the full query set, adding per-query fault isolation so unsupported SQL logs as a coverage result instead of aborting the run.
- Built the Python generation pipeline turning **DuckDB** dsdgen output into row-group-aligned Parquet partitions, the correctness invariant the zip operator requires, plus a verification pass proving reconstructed tables match the source.

Harvard Medical School & Brigham and Women's Hospital

Boston, MA

ML & Data Analytics Intern, Ann Romney Center for Neurologic Diseases

July 2024 – Aug. 2024

- Built a Python pipeline automating RNA/DNA processing and flow-cytometry integration, cutting manual processing time **70%**.
- Trained scikit-learn classifiers on dendritic- and T-cell data to identify multiple sclerosis biomarkers, improving classification **20%**.

Stanford University, Kevin Alexander Lab

Stanford, CA

Data Analytics & Research Intern, Cardiovascular Institute

June 2024 – Aug. 2024

- Engineered **FibroQuant**, a Java and ImageJ macro tool for biomedical image analysis, processing **300+ samples** at **2x** prior throughput.
- Designed thresholding and pattern-recognition algorithms for fibrosis classification ($p < 0.01$), lifting diagnostic throughput **35%**.

University of Maryland, MIND Lab

College Park, MD

Software Engineering & Data Science Intern

June 2022 – Aug. 2023

- Deployed a Python, scikit-learn, and pandas pipeline classifying user state from multimodal respiratory and speech data, published in a peer-reviewed research paper; automated labeling of **450+** breathing segments, cutting annotation time **99%**.

Projects

Do Language Models Say What They Think? | Python, Transformers, LLM-as-Judge

github.com/aarushkandukoori/AI-Research-CoT-Project

- Replicated Turpin et al. (2023) and Anthropic's chain-of-thought faithfulness method: inject biasing hints into BIG-Bench Hard prompts, isolate the answer flips those hints cause, and measure how often the stated reasoning admits the hint.
- Built the pipeline end to end: hint injection, influence detection, an LLM judge validated against 30+ human labels, and 10,000-resample bootstrap CIs, comparing Qwen2.5-1.5B against 7B in 4-bit; shipped an interactive results site on GitHub Pages.

NBA Referee-Crew Market Backtest | Python, LightGBM, Walk-Forward CV

github.com/aarushkandukoori/nba-referee-market-edge

- Tested whether referee-crew tendencies create a tradable edge on game totals over **5,041** training and **1,838** out-of-sample games, using rolling crew features, shrinkage toward a time-varying league prior, and the market line as a control.
- Reported an honest null (log-loss delta of +0.00038, permutation $p = 0.57$), with look-ahead bias designed out via season walk-forward splits, pre-tip-only pricing, and a dedicated leakage test suite.

Publications & Patents

- Comparative Analysis of Digital Tools and Traditional Teaching Methods in Educational Effectiveness.* arxiv.org/abs/2408.06689
- U.S. Provisional Patent #63/686,176: wearable closed-loop atropine delivery system for crashing bradycardia (BUMP).

Technical Skills

- Languages:** Python, Rust, Java, C++, SQL, TypeScript, JavaScript, R
- ML & Data:** PyTorch, ONNX Runtime, scikit-learn, LightGBM, NumPy, pandas, Hugging Face Transformers, LLM evaluation & benchmarking
- Systems & Web:** Docker, Redis Streams, TimescaleDB, Prometheus, GitHub Actions, DataFusion, DuckDB, FastAPI, React, Next.js, AWS, Git